Goodness of Fit Test and QQ Plots

IEOR E4709: Statistic Analysis and Time Series
©2025 by Agostino Capponi

1 p-Values

The p value is the probability that the test statistic takes the value we observed or more extreme values away from the null hypothesis, if the null hypothesis is true. Recall the example from last class, in which the rejection region is

$$C_1 = \left\{ \frac{|\bar{x} - \theta_0|}{s/\sqrt{n}} \geq c \right\},$$

where $\bar{x}$ is the sample mean and s is the sample standard deviation. Suppose $n = 130$, and we use a significance level $\alpha = 0.05$. Because the above is a two-sided test, we should reject the null hypothesis if our test statistic exceeds $t_{n-1,1-\alpha/2} = t_{129,0.975}$ in absolute value. Here, $t_{n-1,1-\alpha/2}$ is the $1-\alpha/2$ quantile of a t distribution with $n-1$ degrees of freedom. We have that $t_{129,0.975} \approx 1.98$. A problem with the critical value method is the following. Suppose the test statistic happens to be $t = 1.94$. Then the conclusion of the test would be to not reject the null hypothesis at $\alpha = 0.05$. However, a value of $t = 1.94$ is right on the border of the rejection region and the evidence is still fairly strong for rejecting the null hypothesis, but not quite at the $\alpha = 0.05$ level. If we test at $\alpha = 0.05$, all we can state is that we fail to reject the null hypothesis at $\alpha = 0.05$ without saying anything on the fact that the actual evidence against the null hypothesis is quite strong. Could we have rejected it using $\alpha = 0.075$ or $\alpha = 0.1$? *What is the smallest significance level at which we can still reject the null hypothesis?* The answer is the **p-value**. The p-value is the smallest significance level at which we can reject the null hypothesis. For this reason, the p-value is also called the *observed significance level*.

2 Goodness of Fit Tests

These tests are used to test the null hypothesis that the data comes from a specific distribution, and they are based on a comparison of the theoretical and the empirical distribution.

Example: In practice, it is very difficult to estimate the probability distribution of defaults (bankruptcy events). Moody's KMV has defined the concept of distance to default to capture the likelihood of a firm's default. This is defined as the distance of the asset value of a firm from its liabilities threshold. If the asset value of the firm falls below its liability threshold, then the firm is unable to cover its debt payments, and therefore will fail. Consider the following procedure. Create

Distance to defaults	DD (0-0.1M)	DD(0.1M-1M)	DD(1M-1.5M)	DD(1.5M,2M)	$DD > 2M$
frequency	9	9	10	14	6

Table 1: Distance to default buckets (M stands for million dollars). The table report number of firms whose distance to default falls in that bucket.

different buckets, each associated with a value for the *distance to default*. Then, for each firm i calculate the distance to default DD (this can be defined, for instance, as the difference between the asset value of firm i, and the firm i's liabilities.) Suppose that this procedure produces the output in Table 1.

The objective is to estimate the probability distribution of the distance to default. Let X_j be the random variable designating in which bucket the distance to default of the j-th firm falls. Assume that the X_j's are independent. To place this assessment in terms of a formal hypothesis testing, let $\pi = (\pi_1, \pi_2, \ldots, \pi_k)$ be postulated values of the probability

$$\mathbb{P}_\pi(\text{a firm is member of the i-th bucket}) = \pi_i.$$

Let the parameter space be $\Theta = \left\{p = (p_1, p_2, \ldots, p_k); p_i \geq 0, \sum_{i=1}^k p_i = 1\right\}$, where k is the number of buckets. This parameter space has $k-1$ free parameters. Once these are chosen, the remaining parameter is determined by the constraint that the sum of the p_i's must equal one. Hence, $dim(\Theta) = k-1$. The hypothesis test is: $H_0 : p_i = \pi_i$ for all i versus $H_1 : p_i \neq \pi_i$ for some i. We have that $dim(\Theta_0) = 0$ because this is a single point in the k-simplex.

Let us use $n_i = \#\{X_j : X_j = i\}$. Hence, the vector $n = (n_1, n_2, \ldots, n_k)$ is a vector that records the number of observations in each bucket. Let $N = \sum_{j=1}^k n_j$ be the total number of observations. The likelihood function $L(p|n) = \frac{N!}{n_1!n_2!\ldots n_k!} p_1^{n_1} p_2^{n_2} \cdots p_k^{n_k}$. The likelihood ratio is the ratio of the maximum value of the likelihood function under the null hypothesis, and the maximum likelihood for any parameter value. Hence, the test statistic is

$$\lambda(n) = \frac{L(\pi|n)}{L(\hat{p}|n)} = \frac{\pi_1^{n_1}\pi_2^{n_2}\cdots\pi_k^{n_k}}{\hat{p}_1^{n_1}\hat{p}_2^{n_2}\cdots\hat{p}_k^{n_k}} = \left(\frac{\pi_1}{\hat{p}_1}\right)^{n_1}\left(\frac{\pi_2}{\hat{p}_2}\right)^{n_2}\cdots\left(\frac{\pi_k}{\hat{p}_k}\right)^{n_k},$$

where $\hat{p}$ is the maximum likelihood estimator, i.e., the maximizer of the likelihood function $L(p|n)$.

To find the MLE, we need to maximize the log-likelihood, $LL(p|n) = \log\left(\frac{N!}{n_1!n_2!\ldots n_k!}\right) + \sum_{i=1}^k n_i \log p_i$, subject to the constraint that $\sum_{i=1}^k p_i = 1$. Using Lagrangian multipliers (verify yourself), it can be checked that $\hat{p_i} = \frac{n_i}{N}$. This is in line with what we would expect intuitively. Hence,

$$\lambda(n) = \left(\frac{N\pi_1}{n_1}\right)^{n_1}\left(\frac{N\pi_2}{n_2}\right)^{n_2}\cdots\left(\frac{N\pi_k}{n_k}\right)^{n_k},$$

We reject the null hypothesis if this ratio is too low, i.e, the likelihood under the null hypothesis is sufficiently smaller than the maximum likelihood under the alternative hypothesis. Using an argument similar to what we did for the likelihood ratio tests in the previous lecture, we can show that, under the null hypothesis H_0, $-2\log\lambda(n)$ has approximately a chi-squared distribution with a

number of degrees of freedom equal to $dim(\Theta) - dim(\Theta_0)$, that is a χ^2_{k-1}-distribution in our setting. We have that

$$-2\log\lambda(n) = -2\sum_{i=1}^{k} n_i \log\frac{N\pi_i}{n_i} = 2\sum_{i=1}^{k} n_i \log\frac{n_i}{N\pi_i}.$$

The above statistics is usually expressed (in textbooks or articles) using $O_i = n_i$ for the number of observed occurrences of i, and $E_i = N\pi_i$ for the number of expected occurrences of i under the null hypothesis H_0. Hence, we can rewrite it as

$$-2\log\lambda(O) = 2\sum_{i=1}^{k} O_i \log\frac{O_i}{E_i}.$$

The expression above is called the G test statistic. Because the rejection rule is to reject the null hypothesis when the test statistic $\lambda(n) \leq c$, this means that we would reject when $-2\log\lambda(n) \geq c_1$. Because $-2\log\lambda(n)$ follows a χ^2_{k-1} random variable, we have that the p-value is the probability that a χ^2_{k-1} distributed random variable takes a value larger or equal than $-2\log\lambda(O)$.

Another statistic used for goodness of fit tests is the Pearson's statistic. This is defined by

$$\Lambda'(O) = \sum_{j=1}^{k} \frac{(E_j - O_j)^2}{E_j}$$

Next, we show the relation between the Pearson's statistic and the G statistic. First, it can be verified that the Taylor expansion of the function $x\log(x/x_0)$ around x_0 is given by

$$x\log(x/x_0) \approx x - x_0 + \frac{1}{2}\frac{(x-x_0)^2}{x_0}.$$

Then

$$\begin{aligned} -2\log\lambda(O) &= 2\sum_{i=1}^{k} O_i \log\frac{O_i}{E_i} \approx 2\sum_{i=1}^{k}\left(O_i - E_i + \frac{1}{2}\frac{(O_i-E_i)^2}{E_i}\right) \\ &= \sum_{i=1}^{k}\frac{(O_i-E_i)^2}{E_i} := \Lambda'(O), \end{aligned}$$

where we have used that $\sum_{i=1}^{k} E_i = \sum_{i=1}^{k} O_i = N$.

The statistics $\Lambda'(O)$ is called the Pearson's statistic. It was introduced between 1895 and 1900 by Karl Pearson, and has been in use for longer than the concept of likelihood ratio tests.

The above argument shows that the Pearson's statistic and the G test statistic are asymptotically equivalent.

Example: Consider Table 1. Assume we want to fit a **Poisson model** to the distribution of the distances to default.[1] Our null hypothesis is $H_0 : p_i = \pi_i$, where $\pi_i(\lambda) = e^{-\lambda}\frac{\lambda^i}{i!}$ for $i = 0, 1, 2, 3,$

[1] We assume each bucket in our data corresponds to an incremental index of the Poisson counts (i.e bucket 0 corresponds to $X = 0$, ..., bucket k corresponds to $X = k$, where X is poisson(λ)).

and $\pi_4 = 1 - \sum_{i=0}^{3} \pi_i$.[2] We obtain $\hat{\lambda} = 1.979$ if we calculate the MLE estimate of the Poisson distribution using the data in Table 1. Next, compute the expected frequency $E_j = N\pi_j(1.979)$, where N is the total number of observations. We get $E_0 = 6.63$, $E_1 = 13.13$, $E_2 = 12.99$, $E_3 = 8.57$, $E_4 = 6.68$. Compute the test statistics $-2\log\lambda(O) = 5.92$, and $\Lambda'(O) = 6.34$. Note that $\Theta = \{p = (p_0, \ldots, p_4) : \sum p_j = 1\}$, and so $dim(\Theta) = 5 - 1$, and $\Theta_0 = \{\lambda : \lambda > 0\}$, so that $dim(\Theta_0) = 1$. Then, both $-2\log\lambda(O)$ and $\Lambda'(O)$ follow a χ^2 distribution with degrees of freedom equal to $5 - 1 - 1 = 3$. The p-values of these two tests can then be readily computed, and it is equal to 0.1155 if we use the statistic $-2\log\lambda(O)$, and 0.0962 if we use the statistic $\Lambda'(O)$. This provides moderate evidence against the null hypothesis that the distance to default follows a Poisson distribution with parameter $\hat{\lambda} = 1.979$.

[2]For practical purposes, in order to apply the G test, we have to account for the fact that buckets are finite, and therefore we choose to truncate the Poisson up to k, and assign the tail distribution to the last bucket. Note that this is not the only way to approach this problem, and it is only a heuristic to proceed with the goodness-of-fit test.